

A Project Report On

MERO DOCTOR: AI FOR PNEUMONIA DETECTION

AND HOSPITAL RECOMMENDATION SYSTEM



By

Aakash Karki	[2021-1-07-0605]
Abiskar Gyawali	[2021-1-07-0607]
Bibas Chhetri	[2021-1-07-0615]
Ram Hari Gyawali	[2021-1-07-0630]
Ritesh Acharya	[2021-1-07-0631]

Submitted to

DEPARTMENT OF COMPUTER ENGINEERING

LUMBINI ENGINEERING, MANAGEMENT AND SCIENCE COLLEGE

Bhalwari

[2025]

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**MERO DOCTOR: AI FOR PNEUMONIA DETECTION
AND HOSPITAL RECOMMENDATION SYSTEM**

In partial fulfilment of requirements for the degree of
Bachelor of Engineering in Computer Engineering



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Ram Hari Gyawali	[2021-1-07-0630]
Ritesh Acharya	[2021-1-07-0631]

Under the Guidance of

Asst. Asso Professor Er.Nirmal Thapa

DEPARTMENT OF COMPUTER ENGINEERING

LUMBINI ENGINEERING MANAGEMENT AND SCIENCE COLLEGE

Bhalwari

[2025]

CERTIFICATE

This is to certify that the project entitled “**MERO DOCTOR: AI FOR PNEUMONIA DETECTION AND HOSPITAL RECOMMENDATION SYSTEM**” has been carried out by a team of 5 members under my supervision in partial fulfillment of the requirements for the degree of Bachelor in Computer Engineering at Pokhara University, Kaski, Nepal.

Team

Aakash Karki (2021-1-07-0605)
Abiskar Gyawali (2021-1-07-0607)
Bibas Chhetri (2021-1-07-0615)
Ram Hari Gyawali (2021-1-07-0630)
Ritesh Acharya (2021-1-07-0631)

Date:

Place: Lumbini Engineering, Management and Science College, Bhalwari

.....
Er. Sunil Kumar Shah
External Examiner

.....
Mr. Satish Kr. Karna
Head of Department
Asst.Asso. Prof.

.....
Er. Nirmal Thapa
Supervisor
Asst.Asso. Prof.

.....
Er. R.K. Shah
Principal
LEMSC

PROJECT APPROVAL CERTIFICATE

This is to certify that the project entitled “**MERO DOCTOR: AI FOR PNEUMONIA DETECTION AND HOSPITAL RECOMMENDATION SYSTEM** ” submitted by the following students:

Abiskar Gyawali (2021-1-07-0605)

Aakash Karki (2021-1-07-0607)

Bibas Chhetri (2021-1-07-0607)

Ram Hari Gyawali (2021-1-07-0630)

Ritesh Acharya (2021-1-07-0631)

This project has been carried out under my supervision, in partial fulfillment of the requirements for the degree of Bachelor in Computer Engineering at Pokhara University, Nepal. The project embodies the original work of the students and has not been submitted earlier for the award of any degree to this institution.

We hereby approve the project work as part of the academic curriculum.

.....

Er. Nirmal Thapa

Supervisor

Asst.Asso. Prof.

.....

Mr. Satish Kr. Karna

Head of Department

Asst.Asso. Prof.

.....

Er. R.K. Shah

Principal

LEMSC

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Aakash Karki

Abiskar Gyawali

Bibas Chhetri

Ram Hari Gyawali

Ritesh Acharya

ABSTRACT

Pneumonia remains a critical global health challenge, causing approximately 2.5 million deaths annually, particularly among children and the elderly. In Nepal, limited access to skilled radiologists and advanced diagnostic tools, especially in rural areas, leads to delayed or inaccurate diagnoses. Mero Doctor is an AI-powered mobile healthcare system that detects pneumonia from chest X-ray images using Convolutional Neural Networks (CNNs), specifically the DenseNet121 model with visual explainability features like heatmaps for high accuracy. After detection, the system recommends nearby hospitals using the K-Nearest Neighbors (KNN) algorithm based on the user's location. Users can search for doctors, view comprehensive doctor lists, check real-time appointment availability, and book appointments with registered doctors through a dedicated system, featuring secure and seamless payment integration via eSewa. Additional features include an AI-powered chatbot using Google Dialogflow for user and doctor queries, and role-based dashboards for Patients, Doctors, and Admins, including managing specializations, doctor verification, feedback, and patient lists. Doctors can manage appointments, mark visited appointments, publish health-related blogs, and interact with users through comments, while patients and doctors can like and comment on blogs to promote health education. Developed using Flutter, Mero Doctor integrates seamlessly into healthcare workflows, enhancing diagnostic accuracy, supporting clinical decision-making, and improving healthcare access in resource-constrained settings like rural Nepal.

Keywords: *Pneumonia Detection, Convolutional Neural Networks (CNN), DenseNet121, Chest X Rays, Hospital Recommendation, Machine Learning, Chatbot, Google Dialogflow*

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List of abbreviations

AI	-	Artificial Intelligence
API	-	Application Programming Interface
CNN	-	Convolutional Neural Network
CXR	-	Chest X-ray
Grad-CAM	-	Gradient-weighted Class Activation Mapping
LSTM	-	Long Short-Term Memory
ML	-	Machine Learning
OpenCV	-	Open-Source Computer Vision Library
ORM	-	Object-Relational Mapping
ReLU	-	Rectified Linear Unit
UI/UX	-	User Interface/User Experience
UML	-	Unified Modeling Language
WHO	-	World Health Organization

CHAPTER-1

INTRODUCTION

1. INTRODUCTION

1.1. Background or Overview

Pneumonia is a serious respiratory infection affecting the lungs, caused by bacteria, viruses, or fungi. It remains a leading global cause of death, especially among children under 5, the elderly, and those with weakened immune systems. According to the WHO, it accounts for about 15% of child deaths under five, killing nearly 800,000 annually. Early detection and prompt treatment are vital for improving survival rates.

Traditionally, radiologists diagnose pneumonia via chest X-rays, a process that is slow, expertise-dependent, and error-prone. In low-resource settings like rural Nepal, the lack of specialists significantly delays diagnosis. Advances in AI, particularly convolutional neural networks (CNNs), provide a solution with fast, accurate image analysis, aiding clinicians with heatmaps and probability scores.

“Mero Doctor” uses CNNs to detect pneumonia from X-rays with high accuracy. It recommends nearby hospitals via the K-Nearest Neighbors (KNN) algorithm only if pneumonia is detected. Users can search doctors, view lists, check availability, and book appointments with secure eSewa payment. Features include a Google Dialogflow chatbot for support and role-based dashboards for Patients, Doctors, and Admins, covering specializations, verification, feedback, and patient lists. Doctors manage appointments, mark visits, publish blogs, and comment, while users like and comment to boost health education. Built with Flutter, it integrates seamlessly into workflows, scaling effectively to underserved regions like Nepal, enhancing diagnostics and patient care.

1.2 PROBLEM STATEMENT

Despite the critical importance of early pneumonia diagnosis, numerous healthcare facilities worldwide, particularly in low-resource settings like rural Nepal, face significant challenges. These include:

1. Lack of specialized radiologists, particularly in rural and remote areas.
2. Limited access to advanced imaging technologies such as digital X-rays and CT scanners.
3. Manual interpretation of chest X-rays is time-consuming and subject to human error.
4. Inconsistent diagnostic accuracy due to varying levels of radiological expertise.
5. Delays in diagnosis and treatment increase the risk of complications or death.
6. Difficulty in accessing timely medical consultations and booking appointments with doctors.
7. Limited availability of nearby hospital information for prompt treatment.
8. Insufficient user engagement tools, such as chatbots or health education platforms, for patient support.
9. Current automated solutions are often too expensive, lack adaptability to local contexts, or do not perform well on non-standard datasets.
10. Mero Doctor addresses these issues by providing an affordable, accurate, and accessible AI-driven diagnostic tool tailored to Nepal's healthcare context.

1.3 OBJECTIVES

The primary objectives of Mero Doctor are:

1. To develop an AI-powered system that accurately detects pneumonia from chest X-ray images using Convolutional Neural Networks (CNNs).
2. To enable users to search for doctors, view doctor lists, and check upcoming appointment availability.
3. To allow users to book appointments with registered doctors, with secure and seamless payment integration via eSewa.
4. To allow doctors to manage appointments, mark visited appointments, publish health-related blogs, and interact with users through comments.
5. To enable patients and doctors to interact by liking and commenting on blogs, promoting health education and engagement.

6. To implement a smart AI-powered chatbot for assisting users and doctors with queries and support.
7. To recommend nearby hospitals based on pneumonia detection results using the K-Nearest Neighbors (KNN) algorithm.
8. To provide role-based dashboards for Patients, Doctors, and Admins with respective functionalities like managing specializations, doctor verification, feedback, and patient lists.

1.4 SCOPE AND APPLICATIONS

1.4.1. Scope of the Project

The project aims to enhance healthcare delivery and outcomes through the following key areas:

1. Develop an AI-powered system to accurately diagnose pneumonia from chest X-ray images using Convolutional Neural Networks (CNNs) with visual explainability features like heatmaps.
2. Improve healthcare accessibility by deploying the system in low-resource and underserved areas, focusing on reducing reliance on radiological expertise.
3. Create a user-friendly mobile interface that seamlessly integrates into healthcare workflows for efficient patient and clinician use.
4. Enable users to search for doctors, view doctor lists, and check upcoming appointment availability to enhance medical access.
5. Allow users to book appointments with registered doctors, incorporating secure and seamless payment integration via eSewa.
6. Enable doctors to manage appointments, mark visited appointments, publish health-related blogs, and interact with users through comments.
7. Facilitate patient and doctor interaction by supporting likes and comments on blogs to promote health education and engagement.
8. Implement a smart AI-powered chatbot to assist users and doctors with queries and support.
9. Recommend nearby hospitals based on pneumonia detection results using the K-Nearest Neighbors (KNN) algorithm.
10. Provide role-based dashboards for Patients, Doctors, and Admins to manage

functionalities such as specializations, doctor verification, feedback, and patient lists.

This approach delivers a comprehensive, scalable solution to transform pneumonia diagnosis and healthcare management.

1.4.2. Applications of the Project

1. Clinical Settings:

- a) Support radiologists and clinicians in hospitals with accurate pneumonia detection and decision-making.
- b) Enable integration into telemedicine platforms for remote diagnostics and consultations.

2. Public Health:

- a) Aid in monitoring pneumonia trends to support timely interventions in underserved regions.
- b) Facilitate screening programs for early pneumonia detection in low-resource areas.

3. Medical Research:

- a) Analyze large datasets to advance understanding of pneumonia patterns and treatments.
- b) Collaborate with healthcare institutions to refine and validate the AI model.

This scope and application framework ensures the project addresses the outlined objectives effectively, catering to diverse healthcare needs.

1.5 Development Methodology

1.5.1 PROCESS MODEL

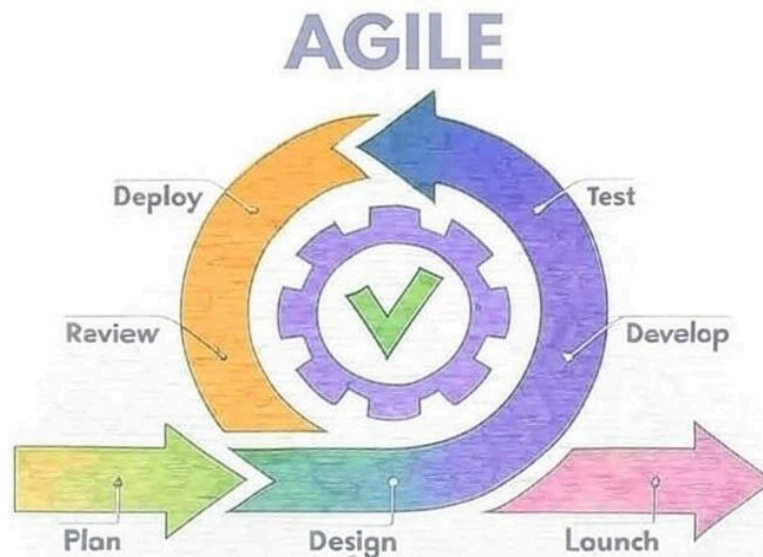


Figure 1.5.1: Agile Development process

The Agile development cycle, a modern software development approach emphasizing flexibility, collaboration, and iterative progress, is ideal for the Mero Doctor project. This model suits the innovative and evolving nature of developing an AI system for pneumonia detection, hospital recommendations, and user engagement features, especially in the context of Nepal's low-resource settings.

The Agile process breaks tasks into iterative sprints, avoiding rigid long-term planning while defining initial project scope, iteration duration, and requirements.

The phases are:

1. **Plan:** Define project requirements, including an AI-powered pneumonia detection system using CNNs with visual explainability (e.g., heatmaps), a chatbot for support, hospital recommendations via KNN, appointment booking with eSewa integration, and role-based dashboards. Collaborate with medical professionals to ensure accuracy and relevance for underserved areas.
2. **Design:** Create the UI for a user-friendly mobile interface, design datasets for CNN training, and plan architectures for chatbot, appointment booking, and dashboard functionalities.

3. **Develop:** Build the CNN model for pneumonia detection, integrate it with the UI, develop the KNN-based hospital recommendation system, implement the chatbot, code appointment booking with eSewa payment, and construct dashboards for Patients, Doctors, and Admins. Include blog features for health education.
4. **Test:** Test the mobile app for bugs, validate pneumonia detection accuracy, ensure eSewa payment security, verify chatbot responses, and confirm dashboard functionalities across roles.
5. **Deploy:** Release the app in a user environment, targeting rural Nepal healthcare settings for accessibility.
6. **Review:** Gather feedback from supervisors, users, and healthcare professionals to refine the model, UI, and features like blogs and appointments.

The Agile model was chosen for its ability to handle machine learning complexities, enabling iterative improvements based on feedback. It supports collaboration among data scientists, developers, and medical experts, ensuring the system meets diverse needs, from diagnostic accuracy to patient engagement, aligning with the project's objectives and scope.

CHAPTER-2

LITERATURE REVIEW

2.1. Description of fundamental theories and general concepts

Pneumonia ranks among the top causes of mortality in children, with a 2022 report from Nepal's Ministry of Health and Population indicating it contributes to about 15% of deaths among children under five. This issue is worsened by geographical isolation, limited availability of modern diagnostic equipment, and a scarcity of trained radiologists, particularly in Nepal's rural and remote regions. Many local health centers and district hospitals struggle with inadequate X-ray interpretation skills, leading to delayed or erroneous diagnoses. In rural medical camps, reliance on clinical symptoms alone often fails to detect early-stage pneumonia accurately.

In recent years, advancements in Artificial Intelligence (AI), especially Machine Learning (ML), have revolutionized medical diagnostics. Research has demonstrated the efficacy of various algorithms in classifying conditions such as cancerous lesions, diabetic retinopathy, pneumonia via CT scans, tuberculosis, and microbial slides. Notably, studies have utilized chest X-rays (CXRs) and CT scans alongside ML models to automate the identification of diverse pneumonia types, including non-COVID-19 viral and bacterial pneumonia, as well as other lung disorders. Our project, "Mero Doctor," builds on these principles by employing Convolutional Neural Networks (CNNs), specifically the DenseNet121 model, to detect pneumonia with high accuracy and visual explainability features like heatmaps. It further integrates the K-Nearest Neighbors (KNN) algorithm to recommend nearby hospitals based on detection results, enhancing timely treatment access. The system also incorporates a user-friendly Flutter-based interface for seamless healthcare integration, enabling doctor searches, appointment bookings with eSewa payment, and an AI-powered chatbot using Google Dialogflow for support. Additionally, role-based dashboards for Patients, Doctors, and Admins facilitate specialized functions like appointment management, blog publishing for health education, and feedback collection, addressing Nepal's healthcare challenges effectively.

2.2. Terminologies related to the project

1. **Pneumonia:** The respiratory infection targeted by the project.
2. **Convolutional Neural Networks (CNNs):** The AI model for pneumonia detection.
3. **DenseNet121:** The specific CNN architecture used.
4. **Visual Explainability Features:** Tools like heatmaps for interpreting AI outputs.
5. **K-Nearest Neighbors (KNN) Algorithm:** The method for hospital recommendations.
6. **Flutter:** The framework for the mobile interface.
7. **eSewa:** The payment integration platform.
8. **Google Dialogflow:** The chatbot technology.
9. **Role-Based Dashboards:** Customized interfaces for different user types.

2.3. Review of the similar/relevant projects

1. **Doctors Hub:** Doctors Hub is an online healthcare service providing affordable virtual consultations and prescriptions via a user-friendly interface. It includes a blog for health education but lacks pneumonia detection.
2. **Qure.ai's qXR:** Qure.ai's qXR is a sophisticated AI-powered platform designed to assist radiologists by automating the analysis of chest X-rays to detect various pulmonary conditions, including pneumonia. It employs advanced convolutional neural networks (CNNs) to analyze X-ray images and identify abnormalities with high accuracy. qXR provides real-time feedback by highlighting areas of concern on the X-ray, which helps radiologists in making quick and informed diagnostic decisions. The system is integrated into radiology workflows, enhancing efficiency and reducing diagnostic turnaround times.
3. **Lunit INSIGHT CXR:** Lunit INSIGHT CXR is an advanced AI tool that analyzes chest X-rays to detect pneumonia and other lung diseases using deep learning algorithms. It provides detailed diagnostic insights, including disease probability, abnormality location, and severity assessments. Designed for seamless integration with existing radiology systems, it improves diagnostic workflows and efficiency. The platform supports radiologists with accurate, actionable data that enhances clinical decision-making.
4. **Aidence - Veye Chest:** Aidence's Veye Chest is an AI diagnostic tool that enhances

chest X-ray interpretation by detecting lung abnormalities, including pneumonia. Using advanced machine learning, it provides real-time analysis and highlights areas of concern. The platform integrates seamlessly with radiology systems and PACS, improving workflow efficiency. Veye Chest continuously learns from new data to maintain diagnostic accuracy.

2.4. Theories and results by other researchers

1. EL. Khalid, Asnaoui, [1] Different types of single and ensemble learning models were utilized to classify pneumonia. Ensemble learning involves combining multiple models into a single model to tackle a specific task, with the choice of models being determined by the requirements and characteristics of the problem at hand. Currently, ensemble models are commonly employed for making predictions, including classification and regression tasks. By training a single model independently within an ensemble, improved accuracy can be achieved. In particular, an ensemble of three models demonstrated higher accuracy in this study.
2. T. Rahman, E.H. Muhammad, [2] used digital x-ray images to detect the bacterial and viral pneumonia. Four different pre-trained deep Convolutional Neural Networks (CNN): AlexNet, ResNet18, DenseNet201, and SqueezeNet were used for transfer learning. This proposed study can be useful in quickly diagnosing pneumonia by the radiologist.
3. P. Pratik, Hem prasad Patil, [3] Early detection and prompt treatment can significantly reduce the mortality rate associated with this disease. Chest x-rays are commonly employed to identify the symptoms of the disease. In this study, a CNN-based model was utilized to automatically detect crucial features, as opposed to using hand-engineered feature selection techniques.
4. Wasif Khan, Nazar Zaki, [4] This helps practitioners to select the most effective and efficient methods from a real-time perspective, review the available datasets, and understand the results obtained in this domain. The usability, goodness aspects, and Computational complexity of the algorithms used for intelligent pneumonia identification are examined in this research.
5. S. Hossain, Rafeed Rahman, [5] The study presented a pneumonia detection approach utilizing MobileNet and ResNet architectures along with Long Term Short Memory (LSTM). This research aims to simplify image analysis for both experts and non-

professionals by proposing a deep learning-based method for pneumonia detection from X-ray images. A dataset containing 5856 X-ray images was utilized, and the maximum accuracy achieved by the proposed LSTM model was noted as 90.2%.

6. D. Meldon, Nimesh Naik, [6] The primary objective of this study is to analyze the patients X-ray images using OpenCV and Deep learning and decide whether the patient has pneumonia or not. We created a deep learning model to aid and address this inconvenient situation for radiologists to achieve the patient's results which can be analyzed and reported to the patient directly. They have used Keras libraries and OpenCV for achieving a high-test data accuracy rate.
7. Abate (2019) et al. [7] mentioned significant achievements of CNNs in large amounts of data sets, but for cases where the data set is small, it sometimes fails if proper care is not taken. They propose a transfer learning approach using pre-trained architectures to obtain the same result between small and large ensembles. They used five pretrained models, where they took the combination of them to form a large ensemble architecture and achieved promising results.
8. Setiawan, et al.[8] (2020) discussed the detection of pneumonia through x-ray images using a convolutional neural network. They worked on the chest X-ray dataset by Kaggle providing an accuracy of 96.30%. X. Li, et al.[9] (2020) presented an improved convolutional neural network for pneumonia detection. Working on the Chest X-ray dataset by Kaggle they were able to get an accuracy of 91.41%.
9. Asnaoui, et al. [10] (2020) displayed a comparison of recently developed deep convolutional neural networks for binary classification of pneumonia images. They were able to achieve an accuracy of 96% on chest X-ray and CT dataset.
10. Antin, et al. [11] (2017) aimed to train a model using the dataset to help physicians in making diagnoses of pneumonia in chest X-ray using a 121 layered convolutional neural network on NIH chest X-ray dataset. Their model could be improved by considering various combinations of features, improving the logistic regression baseline, and including more error analysis.

CHAPTER-3

SYSTEM ANALYSIS

3.1. Requirement Analysis

3.1.1. Functional Requirements:

1. User registration and login with JWT-based role authentication (Patient, Doctor, Admin).
2. Upload and analysis of chest X-ray images for pneumonia detection with visual explainability (e.g., heatmaps) using DenseNet121.
3. Hospital recommendation based on the user's location using KNN algorithm if Pneumonia is detected.
4. Chatbot integration for health queries using Google Dialogflow, supporting users and doctors.
5. Appointment booking and management (request, accept/reject, view status) with secure eSewa payment integration.
6. Doctor profile management with schedule management, blog management, availability functionality.
7. Blog posting by doctors, with patient and doctor interaction via likes and comments for health education.
8. Feedback and rating system for doctors and services.
9. Admin dashboard for managing users, feedback analysis, managing specializations and blogs categories ,verify or reject the doctors.

3.1.2. Non-Functional Requirements:

- 1) Fast performance: Image diagnosis and hospital suggestions within 5 seconds.
- 2) High scalability to handle large datasets and concurrent users.
- 3) Reliable with over 90% pneumonia detection accuracy using DenseNet121.
- 4) Secure data handling with JWT authentication, HTTPS, secure storage, and eSewa payment encryption.
- 5) User-friendly UI built with Flutter, integrating seamlessly into healthcare workflows.
- 6) Maintainable architecture using ASP.NET backend and modular design.
- 7) Continuous learning capability to adapt to new data and feedback.

3.1.3. Hardware Requirements:

1. Development Environment:
 - a) Processor: Intel Core i5 (2.4 GHz or higher).
 - b) RAM: 8GB .
 - c) Storage: 256GB SSD
2. Target Android Device requirement:
 - a) Processor: Quad-core (2.0 GHz minimum).
 - b) RAM: 4GB.
 - c) Storage: 1GB free space (for app installation and caching).

3.1.4. Software Requirements:

1. Development Environment:
 - a) OS: Windows 10/11
 - b) IDE: Visual Studio 2019+ ,Visual Studio Code
 - c) Database: MS-SQL Server 2019 Express/Enterprise.
 - d) Tools: NGROK (for tunneling), Git (version control).
2. Runtime:
 - a) Android: Version 7.0 or later.
 - b) Server: .NET Core 8.0+ runtime

3.2. Feasible Study:

3.2.1. Technical Feasibility

1. **Availability of Tools:** Open-source frameworks like TensorFlow and Keras support CNN development.
2. **Data Sources:** Public datasets (e.g., ChestX-ray14, RSNA) provide training data.
3. **Model Capability:** DenseNet121 achieves >90% accuracy for pneumonia detection.
4. **Deployment:** Cloud or lightweight mobile app deployment for rural accessibility.
5. **Challenges:** Requires quality X-rays and GPU resources for training, lighter models for mobile.

3.2.2. Operational Feasibility:

1. **Ease of Use:** Simple UI with “Positive/Negative” results, hospital recommendations, and blog access.
2. **Integration:** Compatible with healthcare systems and portable X-ray devices, including telemedicine platforms.
3. **Training Requirements:** Minimal training for staff to use UI and interpret heatmaps.
4. **Maintenance:** Regular model and hospital data updates, manageable with a small team.
5. **Acceptance:** High potential in rural areas for faster diagnosis and screening programs.

3.2.3. Economical Feasibility:

1. **Development Cost:** Moderate, reduced by open-source tools and cloud use.
2. **Hardware Requirements:** Basic devices (smartphones) or cloud inference.
3. **Operational Cost:** Limited to cloud fees and updates.
4. **Return on Investment (ROI):** Positive, with cost savings from early detection and NGO support.
5. **Affordability:** Affordable for Nepal with minimal fees, targeting underserved regions.

3.2.4. Schedule Feasibility:

Gantt chart:

<i>Week</i> Topic	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Topic Selection								
Proposal Writing								
Planning								
Designing								
Coding								
Test and Debug								
Report writing								

Figure 3.2.4: Gantt Chart

3.3 Analysis

3.3.1 Data Modeling:

Data modeling is the analysis of data objects and their relationships to other data objects. In this project, different entities like patients, doctors, admin, appointments, blogs, etc are involved. Their relationship can be illustrated through an ER diagram.

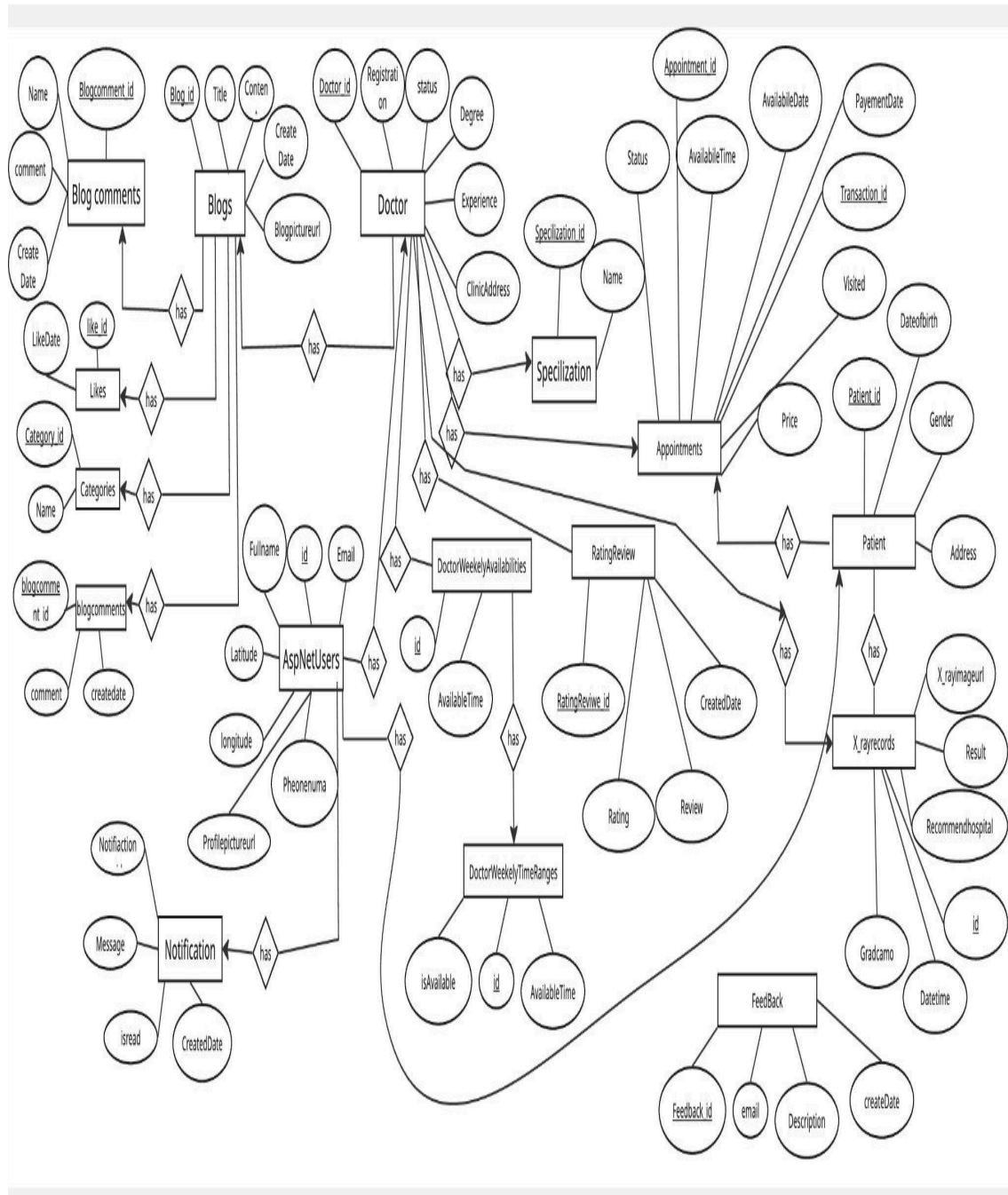


Figure 3.3.1: ER Diagram

3.3.2 Process modeling:

Process modeling refers to the structured representation of a series of activities or tasks that lead to the production of a specific output or service for end users or patients or Doctors. In the context of the Mero Doctor, process modeling is used to visually represent the functional flow of operations that occur within the application.

3.3.3 Data Flow Diagram:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict a good amount of the system requirements graphically. It can be manual, automated, or a combination of both.

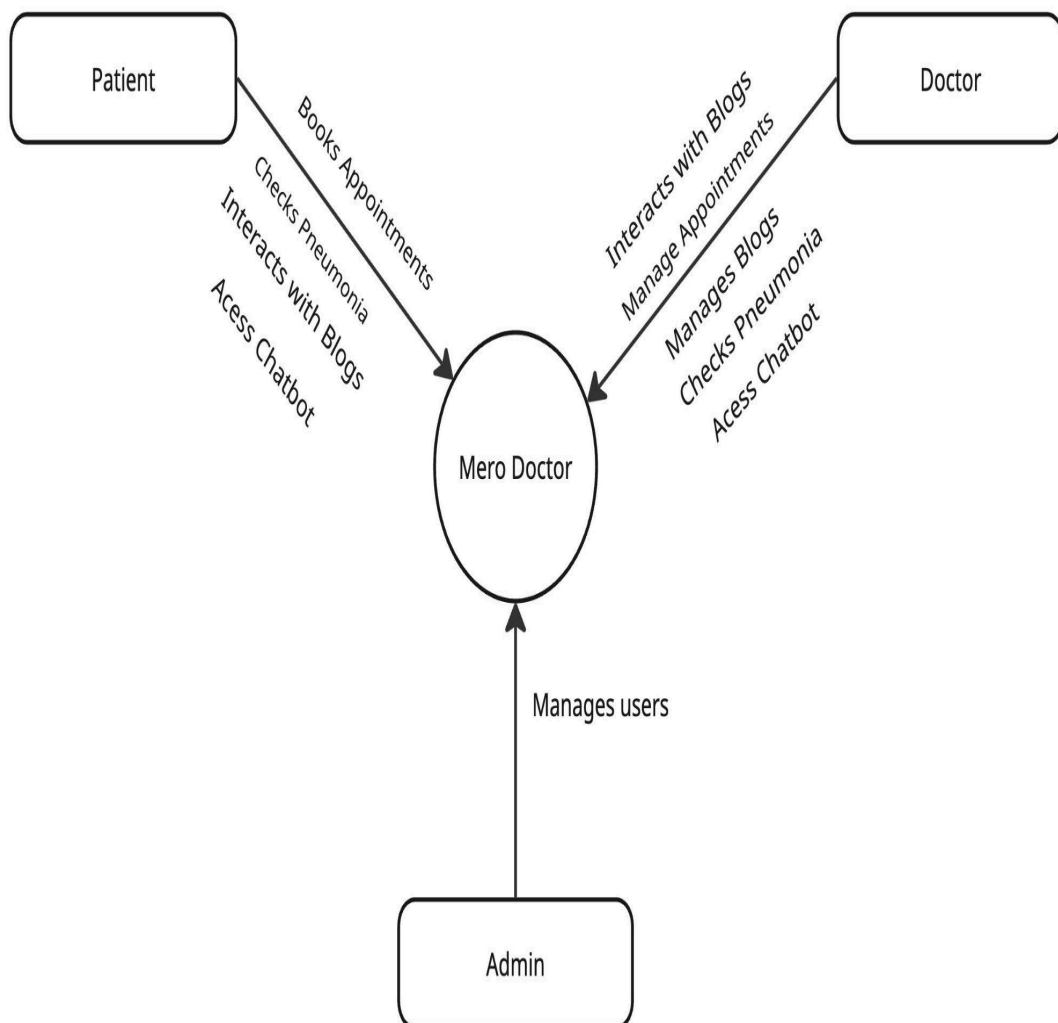


Figure 3.3.3.1: Level 0 DFD of Mero Doctor

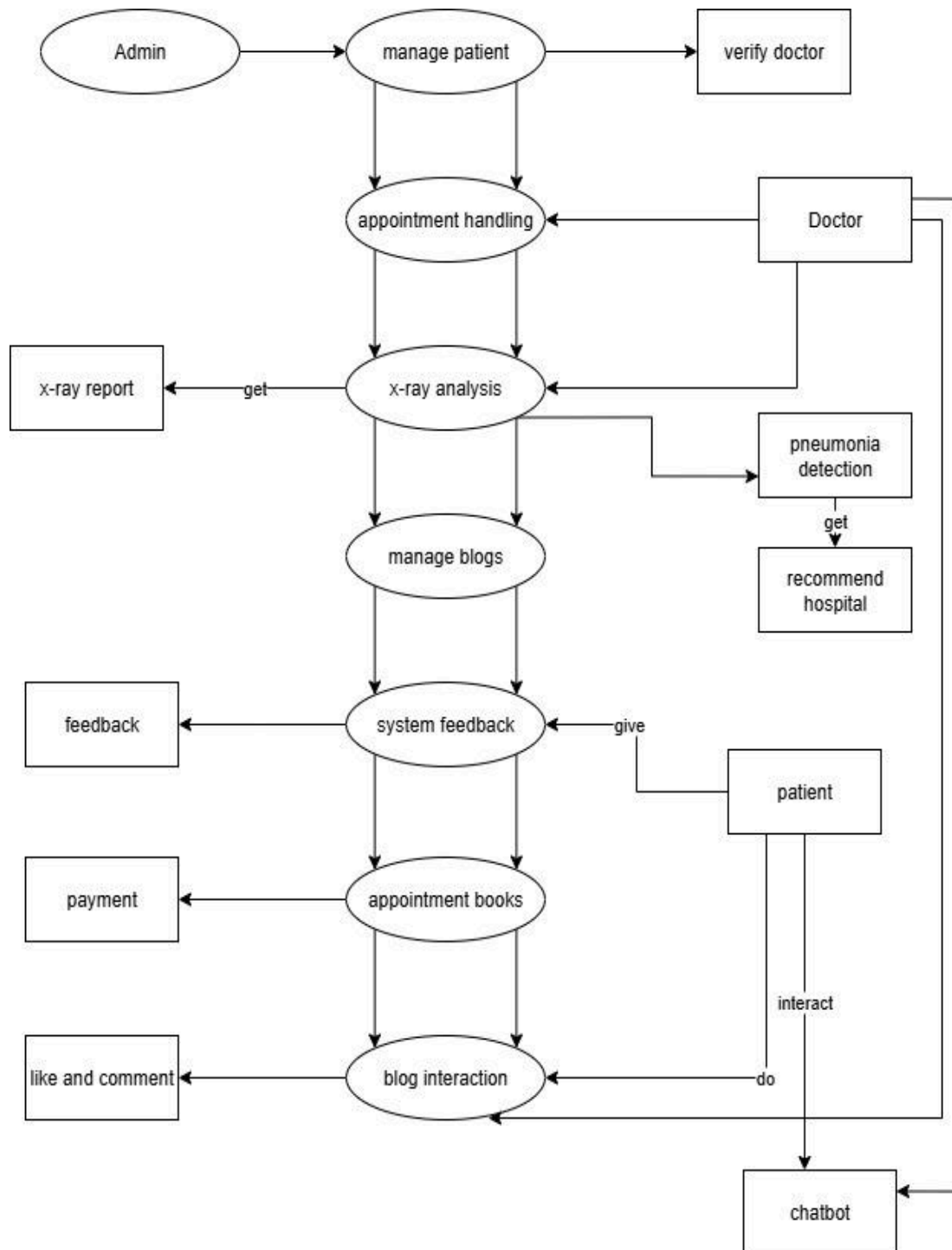


Figure 3.3.3.2: Level 1 DFD of Mero Doctor

Chapter-4: System Design

4.1. System Design

System design is a process of defining the architecture, modules, interfaces and the data for systems to satisfy specified requirements. It is the core concept behind the design of any system. It meets the needs of any organization through an efficient system. Once the requirements are determined, then only the physical system begins to design a good system. System design required to think about everything in an infrastructure from hardware and software

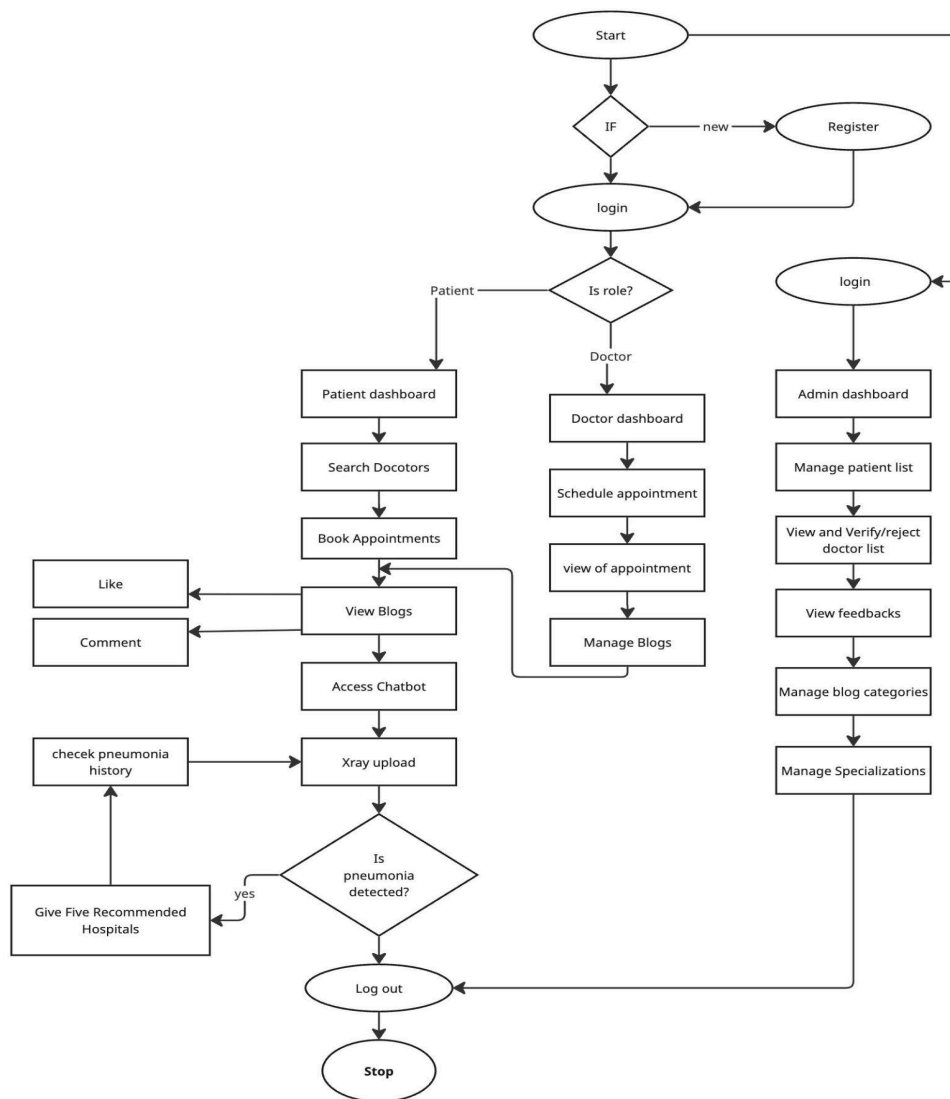


Figure 4.1: Flowchart of System

4.2. Design

4.2.1 Block Diagram

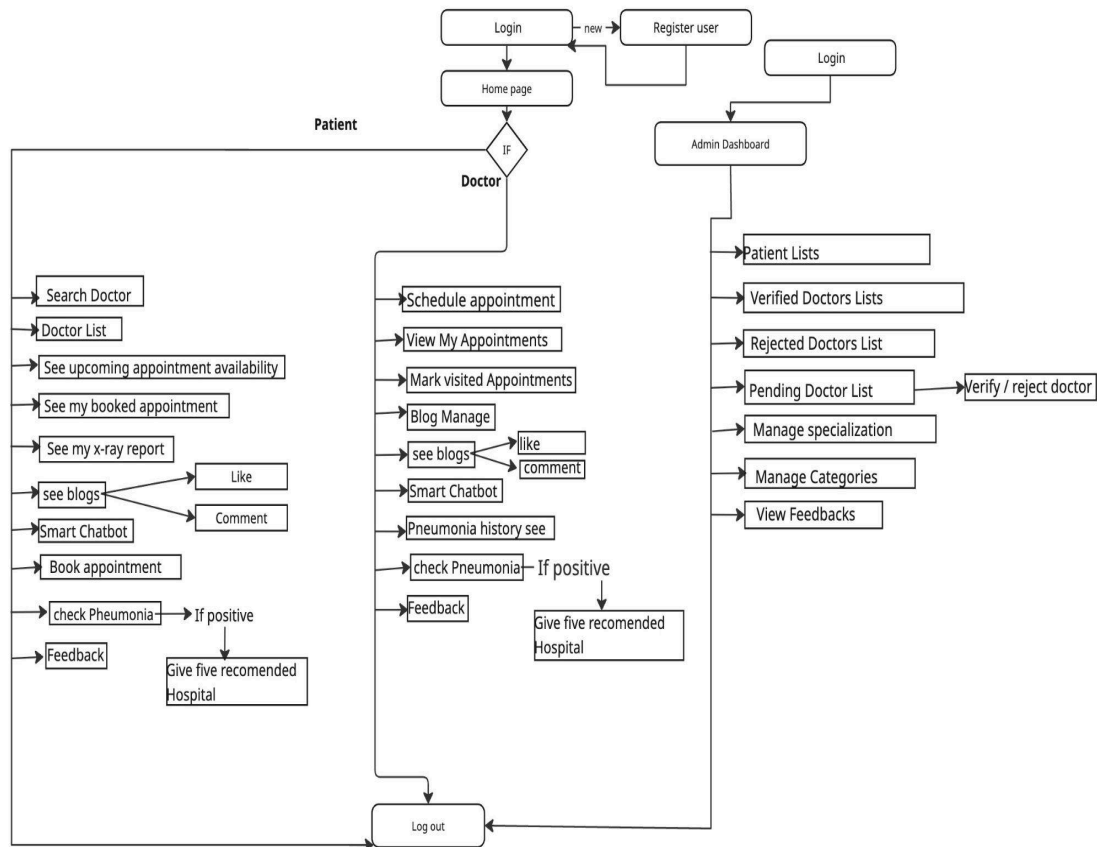


Figure 4.2.1: Block Diagram of Mero Doctor app

4.2.2. Use case Diagram

Use case diagrams are the models that represent the user's expectations to implement the system. The people and the systems that interact with the target system are called actors. The features of the system that the actors are called use cases. Use cases can be written in several ways but the most common way is to represent the system from outside the system.

Use case diagrams are valuable because they:

- Identify the client's expectations of the system.
- Identify the specific features of the system.
- Identify the behavior of the system.
- Provide a simple and easy understood for the clients to view among the features

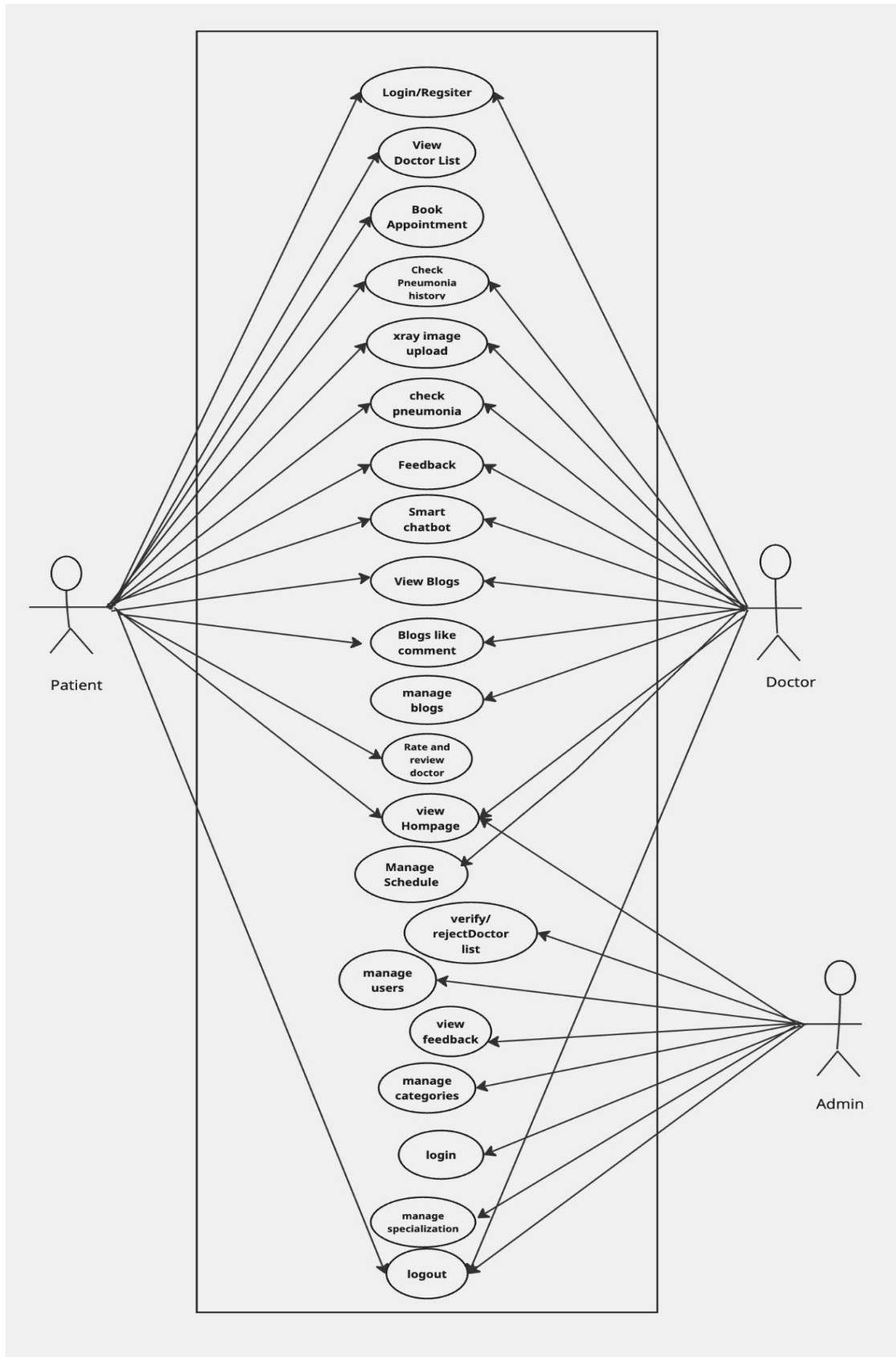


Figure 4.2.2: Use Case Diagram

4.3. Algorithm Used

4.3.1. Convolutional Neural Network (CNN) - DenseNet121:

Description: DenseNet121 is a deep learning CNN architecture that uses dense connectivity to improve feature reuse and efficiency. It is pre-trained on large datasets and fine-tuned for pneumonia detection using transfer learning.

Application: Employed in the Pneumonia Detection module to analyze chest X-ray images (~5,000 from Kaggle dataset). It classifies images as "Pneumonia Positive" or "Negative" with >90% accuracy.

Where Used: Patient and Doctor modules for uploading and analyzing X-rays.

4.3.2. K-Nearest Neighbors (KNN):

Description: KNN is a simple, instance-based ML algorithm that recommends items based on proximity by calculating the distance (e.g., Euclidean) between data points.

Application: Utilized in the Hospital Recommendation module to suggest the top 5 nearest hospitals based on the patient's location (using a dataset of ~30 Nepalese hospitals). It ensures quick access to treatment facilities.

Where Used: Patient module for displaying hospital recommendations.

4.3.3. Google DialogFlow (Natural Language Processing):

Description: DialogFlow is an NLP-based platform that uses machine learning to understand and respond to user queries in natural language.

Application: Powers the AI chatbot to handle pneumonia-related queries, provide system guidance, and support users, enhancing accessibility.

Where Used: Patient and Doctor module for interactive query resolution.

Chapter-5:

Implementation Experiments and Testing

5.1. Implementation

5.1.1. Tools Used

1. IDE (Integrated Development Environment)

An IDE is a software application that provides comprehensive facilities to computer programmers for software development. It typically consists of a source code editor, build automation tools, and a debugger. Its purpose is to maximize developer productivity by providing a central interface for all development tasks.

a) Visual Studio 2022:

A powerful, feature-rich IDE developed by Microsoft. It's primarily used for developing desktop applications (Windows Forms, WPF), mobile apps, cloud services, and games, especially within the Microsoft .NET ecosystem.

It's likely used for developing the ASP.NET backend. It offers excellent debugging tools, code completion (IntelliSense), testing frameworks, and integration with source control and Azure cloud services, making it highly efficient for C# and .NET development.

b) Visual Studio Code:

A lightweight, yet powerful, open-source code editor developed by Microsoft. Unlike a full IDE, it's more flexible and highly extensible via a vast marketplace of extensions, allowing it to support virtually any programming language and framework.

This is the ideal environment for Flutter (Dart) frontend development and Python backend/ML development. VS Code provides excellent support for Dart/Flutter through extensions, enabling hot reload, widget inspection, and debugging. For Python, it offers robust linting, debugging, and environment management. Its versatility makes it a popular choice for multi-language projects.

2. Programming Language

Programming languages are formal languages that specify a set of instructions used to produce various kinds of output. They are used to create programs that control the behavior of a machine, to express algorithms precisely, or as a means of human communication.

1. Frontend:

a) Flutter:

An open-source UI software development kit created by Google. It's used for building natively compiled applications for mobile (Android, iOS), web, and desktop from a single codebase.

b) Dart:

The programming language used by Flutter. It's an object-oriented, client-optimized language for fast apps on any platform.

Choosing Flutter means the project aims for a cross-platform user interface. This allows the team to develop the "MeroDoctor" application and deploy it to mobile (Android/iOS) significantly saving development time and resources while providing a consistent user experience.

2. Backend:

Refers to the "server-side" of an application, which handles data storage, business logic, and communication with the database. Users don't directly interact with the backend; it powers the frontend.

a) ASP.NET:

A free, open-source web framework developed by Microsoft for building web applications and services. It typically uses C# as the programming language. Likely used for building the core web APIs, user authentication, and managing the main application logic, especially if there's a need for a robust, scalable, and secure backend that integrates well with Microsoft technologies like SQL Server and Visual Studio.

b) Python:

A high-level, interpreted, general-purpose programming language known for its

readability and vast libraries. Used for the AI/ML components of the project, specifically for the CNN-based pneumonia detection and possibly the KNN algorithm. Python's extensive ecosystem of ML libraries (TensorFlow, Keras, OpenCV) makes it the industry standard for AI development. It could also handle other backend logic where its data processing capabilities are beneficial.

3. Database

A database is an organized collection of structured information, or data, typically stored electronically in a computer system. It's designed to efficiently store, retrieve, and manage large amounts of information.

a) SQL Server:

A relational database management system (RDBMS) developed by Microsoft. It uses SQL (Structured Query Language) to manage and query data stored in tables. SQL Server is a robust and scalable choice for storing structured data like patient records, doctor information, admin details. Its strong integration with ASP.NET and the Microsoft ecosystem makes it a natural fit for projects using those technologies. It provides features for data integrity, security, and high availability.

4. ML Libraries (Machine Learning Libraries)

ML libraries are collections of pre-written functions, algorithms, and tools that facilitate the development and deployment of machine learning models. They abstract away complex mathematical and computational details, allowing developers to focus on model design and data.

a) TensorFlow:

An open-source machine learning framework developed by Google. It's widely used for developing and training machine learning models, especially deep learning models (like CNNs) for various tasks such as image recognition, natural language processing, and more. CNN-based pneumonia detection is almost certainly built and trained using TensorFlow. It provides the core computational graph engine necessary for handling the complex operations involved in deep neural networks.

b) **Keras:**

A high-level neural networks API, written in Python and capable of running on top of TensorFlow (among other backend engines). Keras is designed for fast experimentation with deep neural networks.

While TensorFlow provides the underlying power, Keras likely offers a more user-friendly interface for building and training the CNN model. It simplifies common deep learning tasks, allowing developers to rapidly prototype and iterate on their pneumonia detection model.

c) **OpenCV:**

(Open Source Computer Vision Library) An open-source library that provides a wide range of tools and algorithms for computer vision tasks, including image and video processing, object detection, and machine learning.

For a project involving chest X-rays, OpenCV is crucial for image processing tasks. This could include loading X-ray images, resizing them, normalizing pixel values, applying filters, or performing other pre-processing steps before they are fed into the CNN model for pneumonia detection.

5. Operating System

The operating system (OS) is system software that manages computer hardware and software resources and provides common services for computer programs.

a) **Windows 10/11:**

Versions of Microsoft's popular desktop operating system. This indicates that the development environment for the "MeroDoctor" project is primarily Windows-based. Developers are using machines running Windows 10 or 11 to write code, run IDEs, train models, and perform testing. While the deployed application might run on other OS (e.g., mobile devices), Windows is where the development work happens.

6. Machine Learning Algorithms:

a) **Convolutional Neural Network (CNN) - DenseNet121:**

Description: DenseNet121 is a deep learning CNN architecture that uses dense connectivity to improve feature reuse and efficiency. It is pre-trained on large datasets and fine-tuned for pneumonia detection using transfer learning.

Application: Employed in the Pneumonia Detection module to analyze chest X-ray images (~5,000 from Kaggle dataset). It classifies images as "Pneumonia Positive" or "Negative" with >90% accuracy. Where Used: Patient and Doctor modules for uploading and analyzing X-rays.

b) K-Nearest Neighbors (KNN):

Description: KNN is a simple, instance-based ML algorithm that recommends items based on proximity by calculating the distance (e.g., Euclidean) between data points.

Application: Utilized in the Hospital Recommendation module to suggest the top 5 nearest hospitals based on the patient's location (using a dataset of ~30 Nepalese hospitals). It ensures quick access to treatment facilities.

Where Used: Patient module for displaying hospital recommendations.

c) Google DialogFlow (Natural Language Processing):

Description: DialogFlow is an NLP-based platform that uses machine learning to understand and respond to user queries in natural language.

Application: Powers the AI chatbot to handle pneumonia-related queries, provide system guidance, and support users, enhancing accessibility.

Where Used: Patient and Doctor module for interactive query resolution.

5.1.2 Implementation Details of Modules

Table 5.1.2 Implementation of Modules

Module	Interface	Purpose	Key Classes / Methods / Functions	Algorithm / Description
Admin	IAdminRepository	Manage doctor verification, admin dashboard, doctor lists	VerifyDoctorAsync(id, status), GetVerifiedDoctorsAsync(), DashboardView()	Update doctor status in DB; Retrieve filtered doctor lists; Provide dashboard summary data
Appointment	IAppointmentRepository	Book, update, retrieve appointments for doctors and patients	BookAppointmentAsync(dto, patientId), UpdateAppointmentStatusAsync(), GetAppointmentsByDoctorAsync()	Validate doctor availability; Book appointment; Update appointment status; Fetch appointments by role and date
Authentication (Doctor)	IAuthDoctorRegistrationRepository	Doctor login and registration	DoctorLoginAsync(model), DoctorRegisterAsync(model)	Authenticate credentials or create new doctor account
Authentication (Patient)	IAuthPatientRegistrationRepository	Patient registration	PatientRegisterAsync(model)	Register new patients
Authentication (General)	IAuthRepository	User login for all roles	LoginAsync(loginDto)	Authenticate user login and issue token
Blog	IBlogRepository	Manage blogs (create, update, delete, retrieve)	AddAsync(dto, userId), UpdateAsync(), GetAllAsync()	CRUD on blog posts; Filter by doctor or user
Blog Comments	IBlogCommentRepository	Manage comments on blogs	AddCommentAsync(), UpdateCommentAsync(), GetCommentsByBlogIdAsync()	Add/update/delete comments with user authorization

Doctor	IDoctorRepository	Retrieve doctor info and apply filters	GetDoctorByIdAsync(), GetDoctorsByFilterAsync(), GetAllTopDoctorsAsync()	Get doctor data filtered by name/specialization; retrieve top-rated doctors
Doctor Weekly Availability	IDoctorWeeklyAvailabilityRepository	Manage availability time slots for doctors	SetDoctorAvailabilityAsync(), GetDoctorAvailabilityAsync()	Add/update/delete weekly available time slots per doctor; fetch upcoming availability
Patient	IPatientRepository	Manage patient information	GetPatientByIdAsync(), GetAllPatientsAsync()	Retrieve patient profiles
Rating & Reviews	IRatingReviewRepository	Allow patients to rate and review doctors	CreateRatingAsync(), UpdateRatingAsync(), GetAllRatingsForDoctorAsync()	Create or update ratings; fetch user rating for a doctor; aggregate ratings
Feedback	IFeedbackRepository	Collect and retrieve patient feedback	CreateFeedbackAsync(), GetAllFeedbacksAsync()	Submit feedback; Retrieve all feedback entries
Likes	ILikeRepository	Toggle likes on blogs/posts	ToggleLikeAsync()	Add or remove like by user and update like count
Specializations	ISpecializationRepository	Manage medical specializations	GetAllAsync(), AddAsync(), UpdateAsync(), DeleteAsync()	CRUD operations for specializations
Pneumonia Detection & X-rays	IXRayRecordRepository	Process X-rays and detect pneumonia, manage X-ray history	DetectPneumonia(IFormFile, userId), GetUserXRayHistory(userId)	Detect pneumonia using ML model (preprocess -> infer -> return result); Maintain user X-ray history

Algorithms

Table 5.1.3: Algorithms

Algorithm Name	Description
predict() (pneumonia detection)	1. Preprocess the X-ray image (resize, normalize). 2. Load pretrained ML model (e.g., CNN). 3. Run inference on image. 4. Output prediction: Pneumonia detected/not detected.
haversine_distance(lat1, lon1, lat2, lon2)	1. Convert degrees to radians. 2. Compute spherical distance formula. 3. Return distance in kilometers.
recommend_hospitals(patient Lat, patientLon, hospitals)	1. Calculate distance using haversine_distance. 2. Sort hospitals by distance ascending. 3. Return top N closest hospitals.

5.2. Testing

To ensure the effectiveness and correctness of the AI for Pneumonia Detection and Hospital Recommendation System, the following levels of testing were conducted:

5.2.1. Test Cases for Unit Testing

Unit Test Cases

Table 5.2.1: Unit test cases

Test ID	Description	Input	Expected Output
UT-001	Verify that user login accepts valid credentials	Valid username and password	Successful login and redirection to dashboard
UT-002	Verify that user login rejects invalid credentials	Invalid username or password	Error message displayed
UT-003	Verify patient registration accepts valid details	Valid patient name, email, password	Registration successful, patient account created
UT-004	Verify registration rejects invalid details	Invalid email or weak password	Error message displayed

UT-005	Verify admin login accepts valid credentials	Valid admin username and password	Successful login and redirection to admin dashboard
UT-006	Verify admin login rejects invalid credentials	Invalid admin username or password	Error message displayed
UT-007	Verify X-ray upload accepts valid image files	Valid X-ray image file	File uploaded successfully and processed
UT-008	Verify X-ray upload rejects invalid file types	Non-image file (e.g., PDF)	Error message displayed
UT-009	Verify pneumonia detection on infected X-ray	X-ray image with pneumonia	True (pneumonia detected)
UT-010	Verify pneumonia detection on healthy X-ray	X-ray image without pneumonia	False (no pneumonia detected)
UT-011	Verify blog upload accepts valid content	Valid blog text and media	Blog uploaded successfully
UT-012	Verify like functionality on a blog post	Click like button on blog	Like count increases by 1
UT-013	Verify comment submission on a blog post	Valid comment text	Comment posted successfully under blog
UT-014	Verify search returns correct doctor results	Doctor name or specialty	List of matching doctors displayed
UT-015	Verify admin can verify a doctor	Valid doctor details	Doctor status updated to verified
UT-016	Verify admin can reject a doctor	Valid doctor details	Doctor status updated to rejected

5.2.2 Test cases for integration testing

Integration Test Cases

Table 5.2.2: Integration test cases

Test ID	Description	Input	Expected Output
IT-001	Verify transition from login to dashboard	Valid user credentials	User redirected to respective dashboard
IT-002	Verify end-to-end patient registration process	Valid patient details	Patient registered and can log in
IT-003	Verify admin login redirects correctly	Valid admin credentials	Admin redirected to admin dashboard
IT-004	Verify X-ray upload triggers detection	Upload X-ray image	Pneumonia detection result displayed
IT-005	Verify end-to-end blog upload by doctor	Doctor logs in, uploads blog	Blog visible in "See Blog" section
IT-006	Verify like and comment features integration	User/doctor likes/comments on blog	Updated like count and comment displayed
IT-007	Verify appointment scheduling process	Select doctor, choose time slot	Appointment scheduled and viewable
IT-008	Verify admin manages/verifies patient list	Add patient, verify existing	Updated patient list with verified status
IT-009	Verify admin can verify or reject doctors	Select doctor, verify/reject	Doctor status updated accordingly

6.1. Conclusion

The development of Mero Doctor has successfully addressed critical healthcare challenges in rural Nepal, where pneumonia remains a leading cause of child mortality due to limited access to radiologists and diagnostic tools. By implementing an AI-powered system using Convolutional Neural Networks (CNNs) with DenseNet121, the project achieves over 90% accuracy in detecting pneumonia from chest X-ray images, enhanced by visual explainability features like heatmaps. The integration of the K-Nearest Neighbors (KNN) algorithm provides effective hospital recommendations, improving timely treatment access. The Flutter-based mobile interface, coupled with secure eSewa payment integration, enables seamless appointment booking and doctor searches, while the Google Dialogflow chatbot and role-based dashboards enhance user engagement and administrative efficiency. Blog features further promote health education through doctor-patient interactions. This comprehensive solution aligns with the project's objectives, significantly reducing diagnostic delays and enhancing healthcare accessibility in underserved regions, as validated through rigorous testing and implementation.

6.2. Future Enhancement / Recommendations

We aim to implement the following improvements in future:

1. Expand the AI model to detect additional lung conditions beyond pneumonia, utilizing transfer learning on diverse datasets.
2. Enhance the chatbot with multilingual support to cater to Nepal's linguistic diversity, improving accessibility.
3. Integrate real-time patient monitoring via wearable devices to provide early warning signs, complementing X-ray analysis.
4. Expand the hospital database with real-time updates to ensure accurate recommendations.

5. Integrate telemedicine for remote consultations to strengthen rural healthcare delivery.
6. Incorporate continuous learning to adapt to new medical data, enhancing model performance over time.
7. Conduct user training programs in rural areas to ensure long-term sustainability and adoption.

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10. A. Asnaoui *et al.*, "Automated methods for detection and classification of pneumonia based on X-ray images using deep learning," *arXiv preprint*, arXiv:2003.14363, 2020.
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APPENDIX



Figure A.1: Welcome page

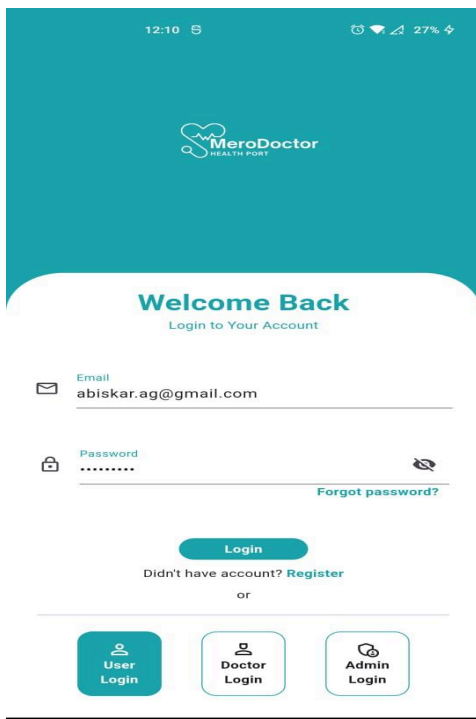


Figure A.2: Patient login/register

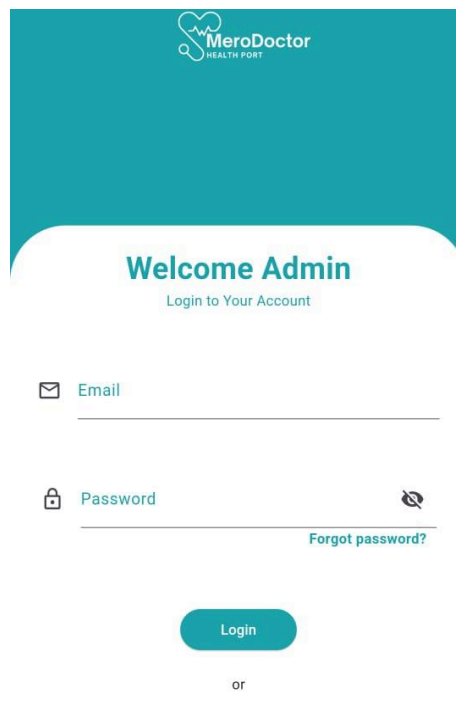


Figure A.3: Admin login

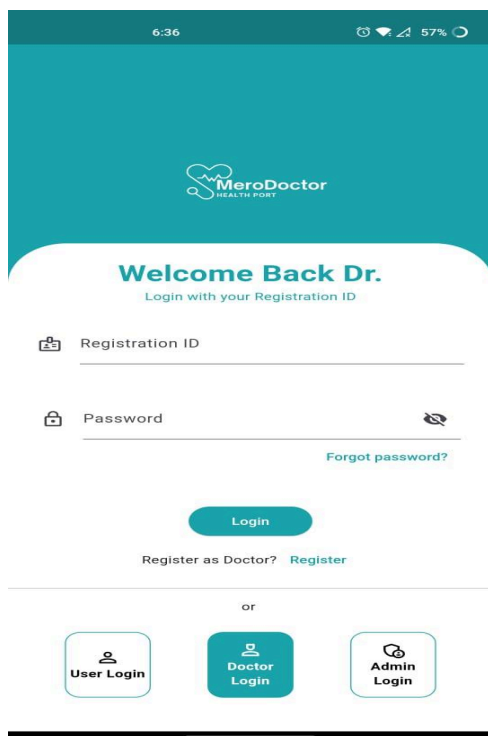


Figure A.4: Doctor login/register

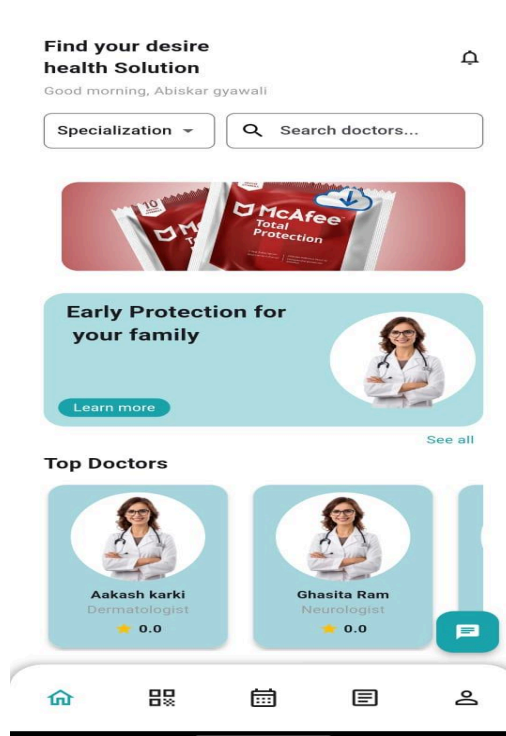


Figure A.5: User homepage

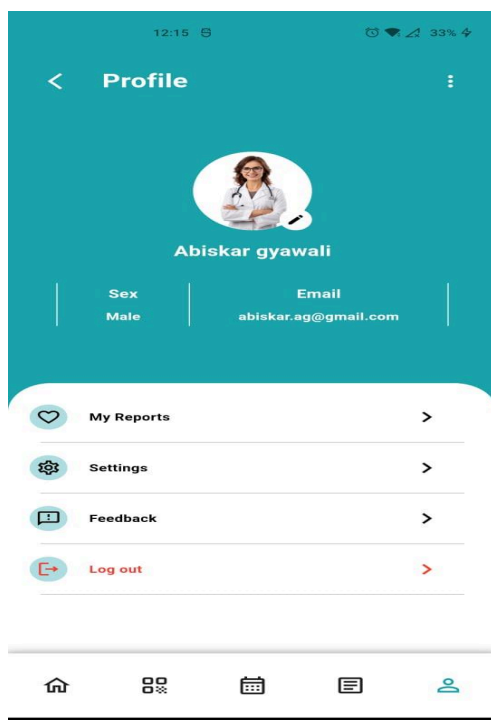


Figure A.6: User profile page

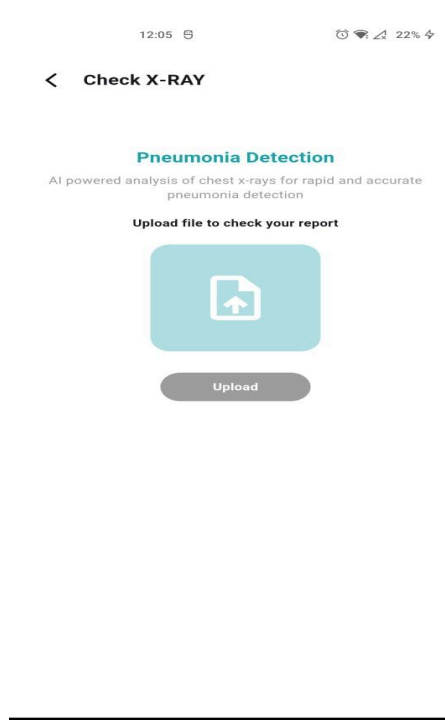


Figure A.7: Check X- Ray page

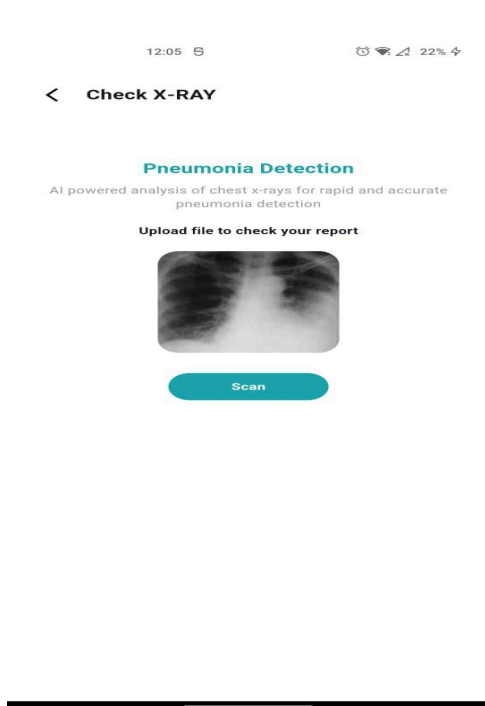


Figure A.8: X-Ray upload and scan

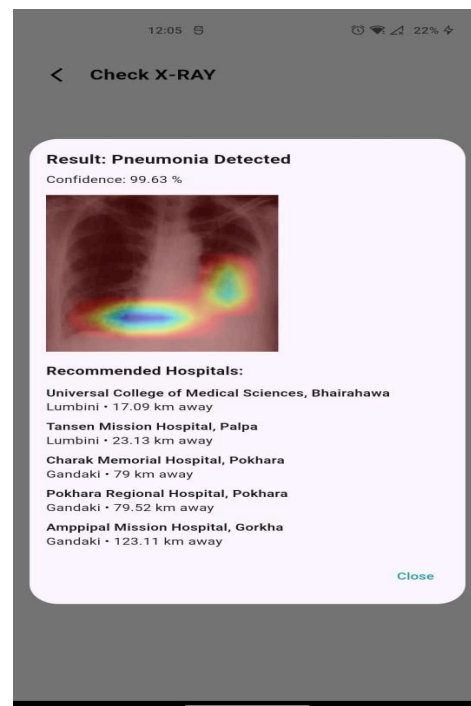


Figure A.9: Pneumonia Report

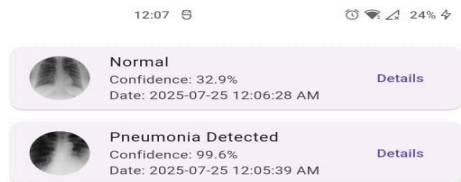


Figure A.10: Patient X-ray test History

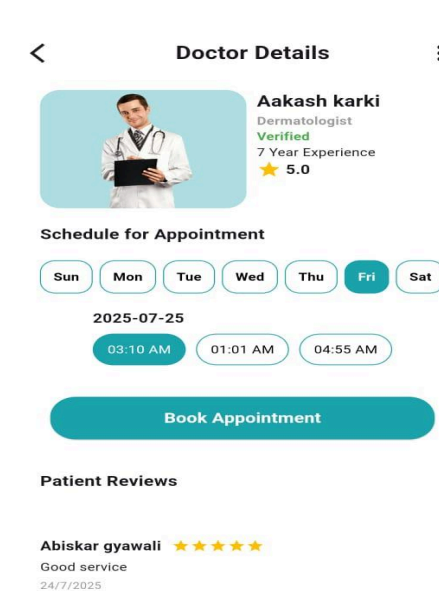
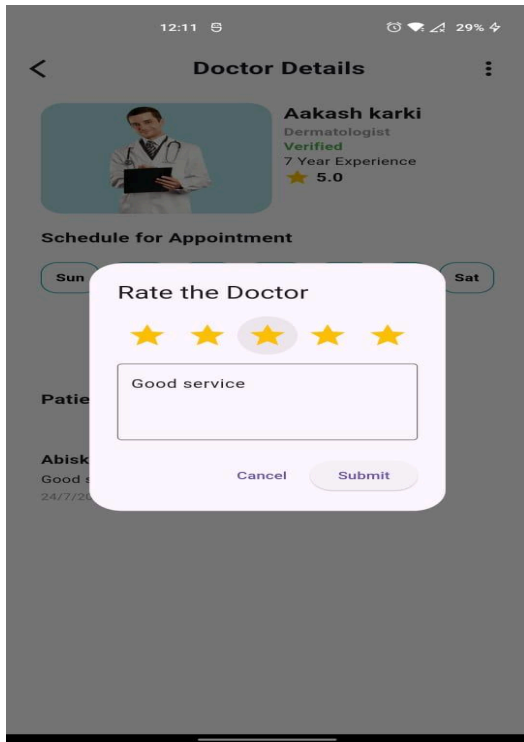


Figure A.11: Doctor Details page



FigureA.12: Patient rate/review to doctor

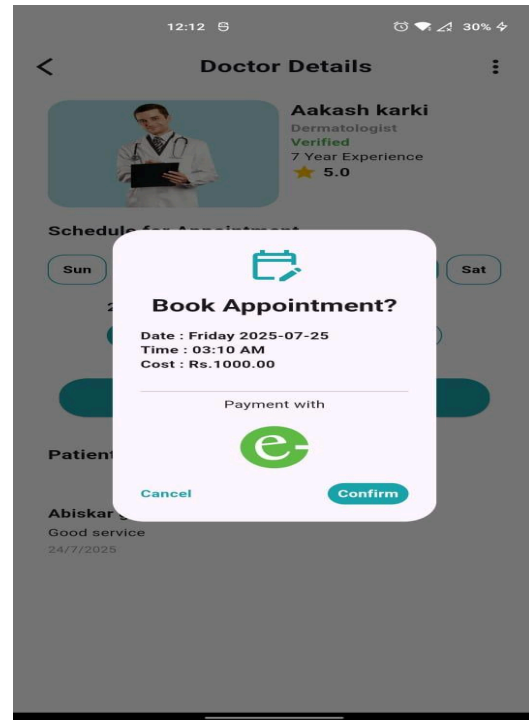


Figure A.13: Doctor appointment page

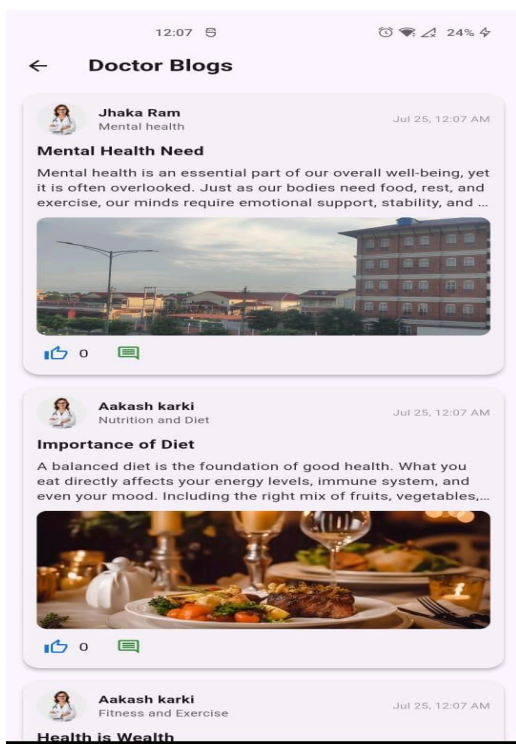


Figure A.14: Blog page

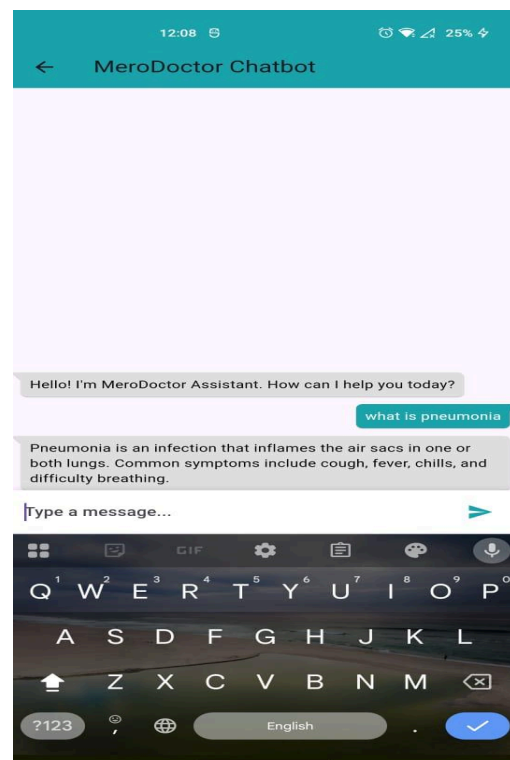


Figure A.15: Chatbot page

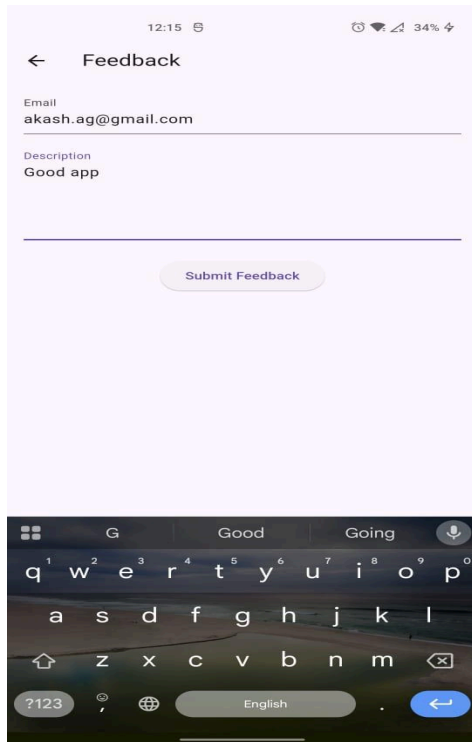


Figure A.16: Feedback Page

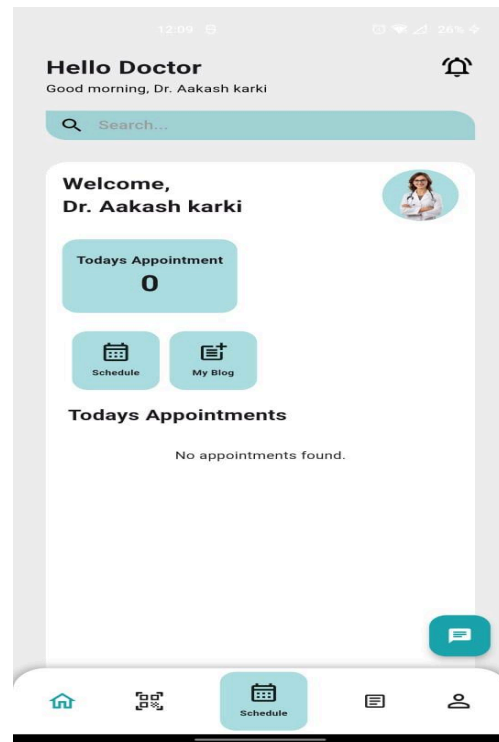


Figure A.17: Doctor Homepage

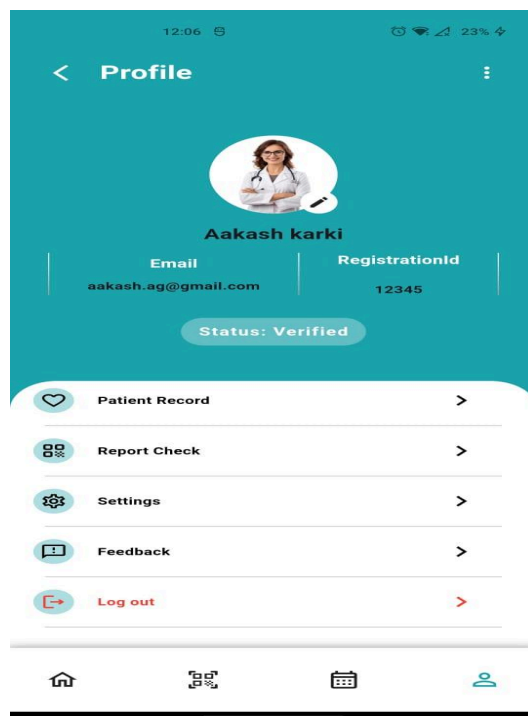


Figure A.18: Doctor Profile

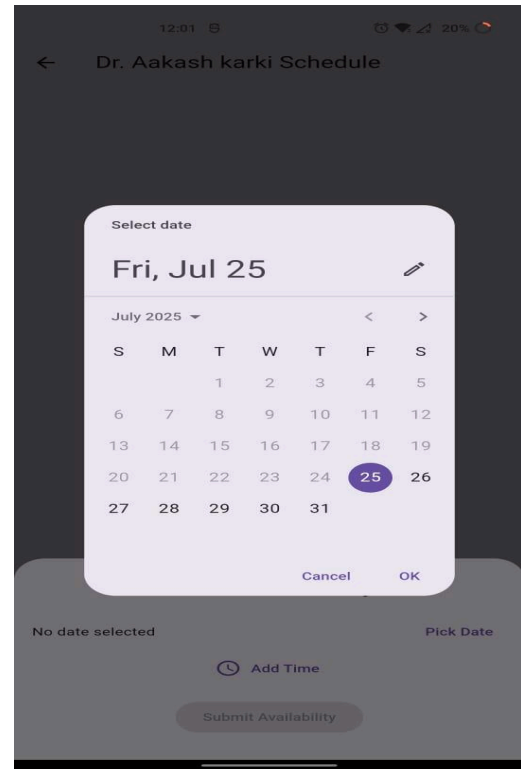
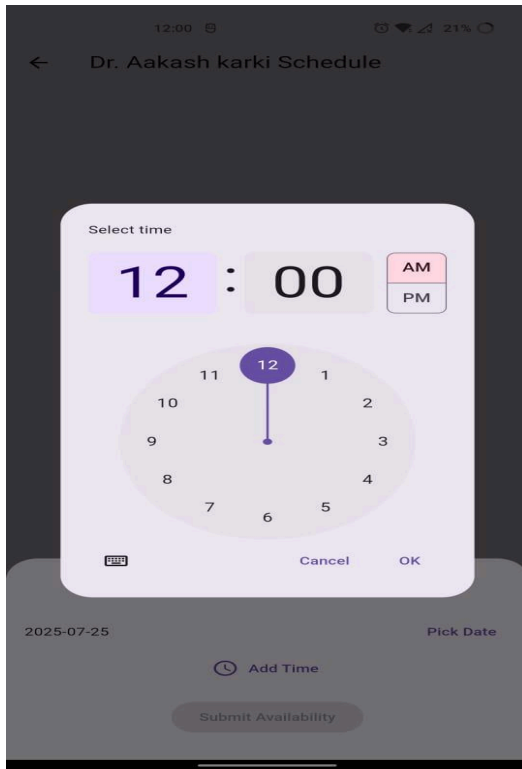


Figure A.19: Doctor schedule for the specific time and date

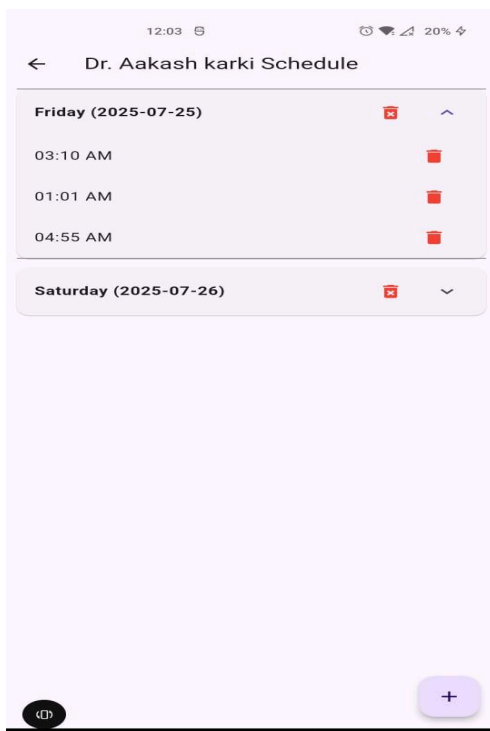


Figure A.20: Doctor Schedule on Saturday

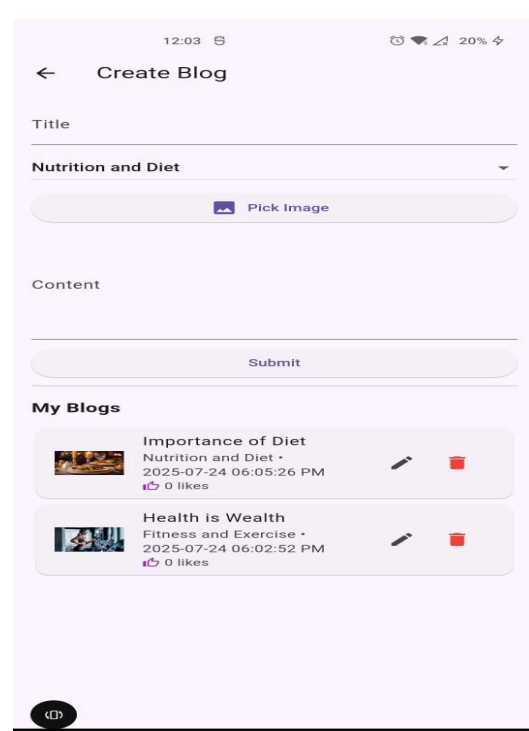


Figure A.21: Blog creation page

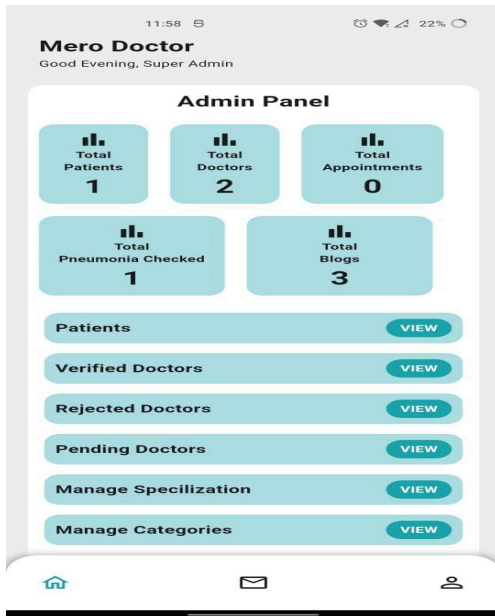


Figure A.22: Admin panel

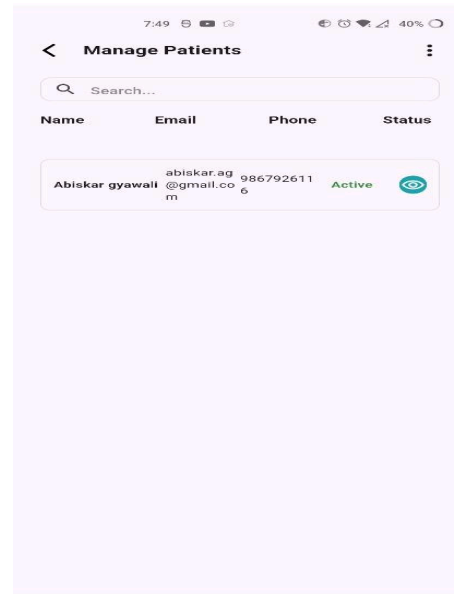


Figure A.23: Patients handling page

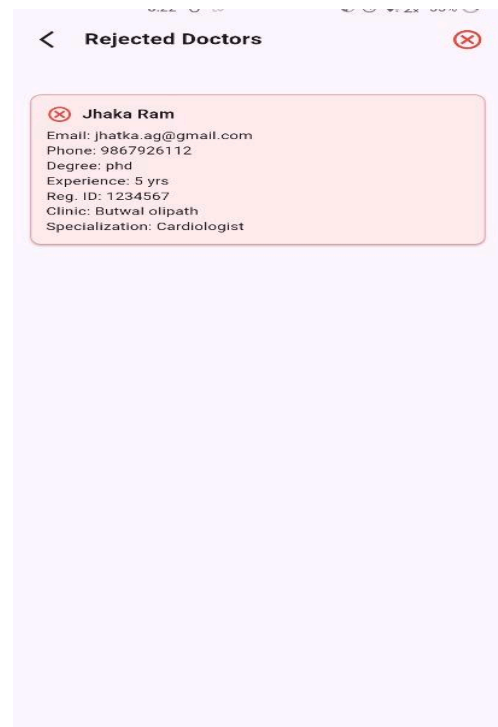


Figure A.24: verification and rejection of Doctors by admin

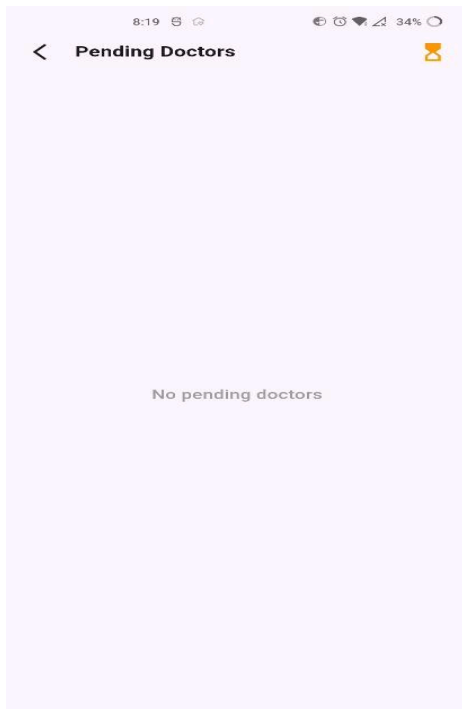


Figure A.25: Doctor Pending

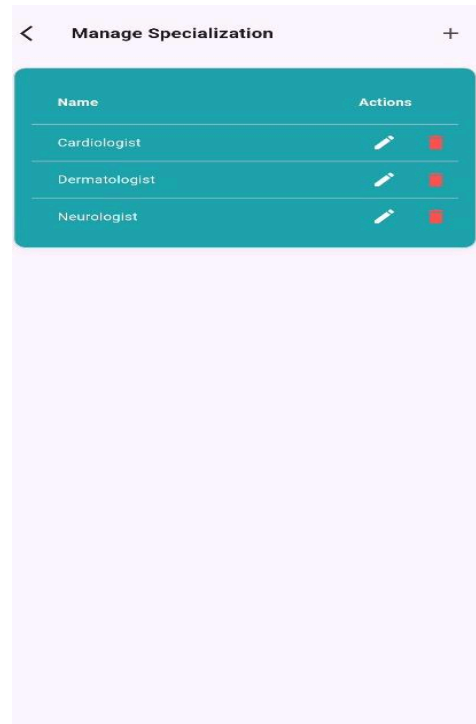


Figure A.26: Specialization

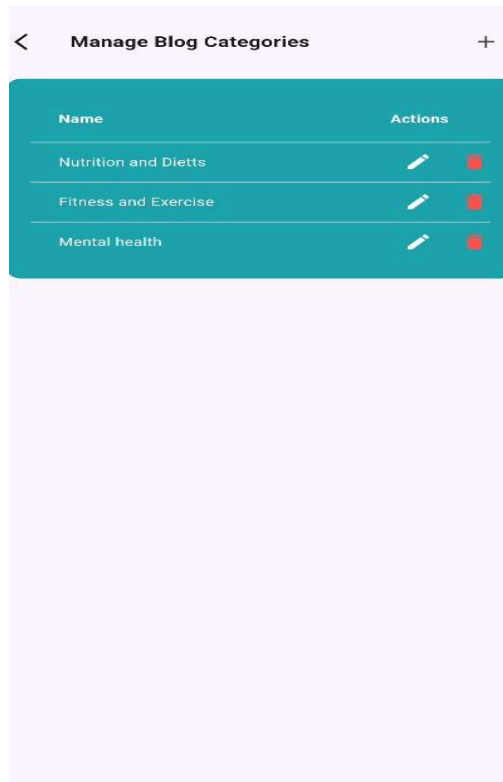


Figure A.27: Blog Categories



Figure A.28: Feedbacks